

AI-Based Onion Quality Inspection & Automated Grading Report
SIH Technical Produce Evaluation • Team: THE DEBUGGERS

1. Executive Batch Inspection Summary

TOTAL ONIONS	QUALITY SCORE	HEALTHY BULBS	DEFECT RATE	AVG SIZE
1	55.0 / 100	1 / 1	0.0%	24.7 mm

2. Prototype Grade Distribution Breakdown

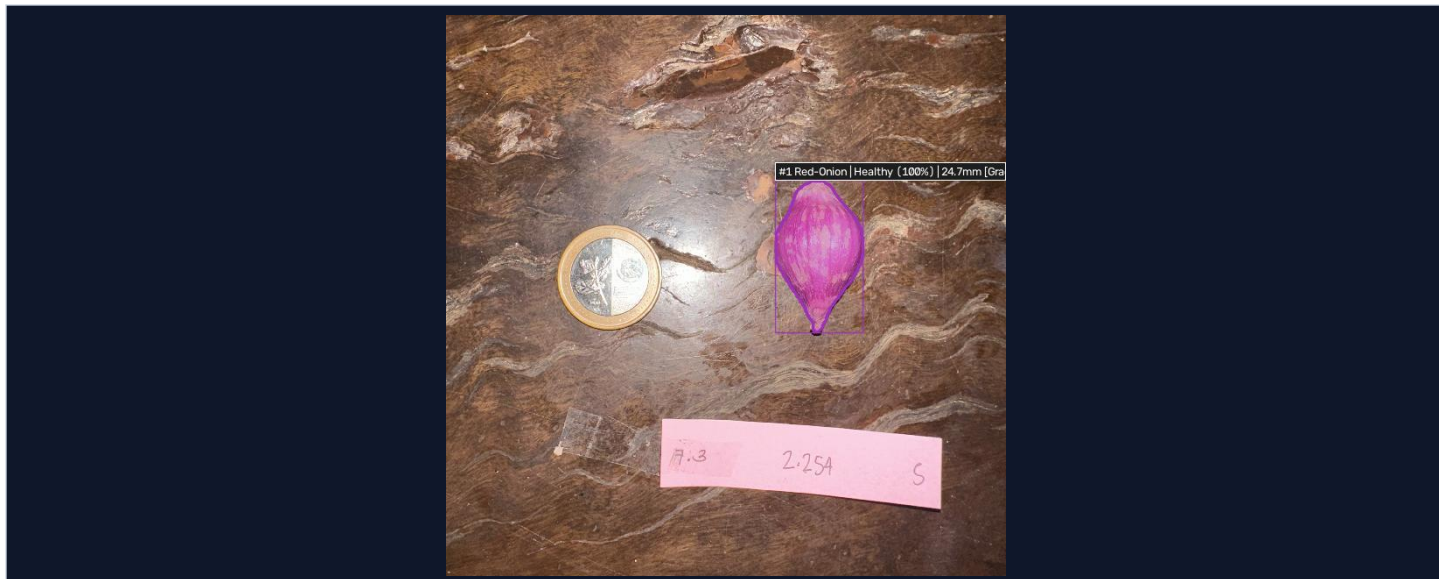
Grade Tier	Count	Share	Commercial Standard Description
Grade A (Premium)	0	0.0%	Sound health, uniform size (45–75 mm when calibrated), negligible surface defect (≤5%).
Grade B (Commercial)	0	0.0%	Good commercial quality, minor size variance (35–45 or 75–90 mm), minor blemishes (≤10%).
Grade C (Marginal)	1	100.0%	Marginal quality, undersized (<35 mm), oversized (>90 mm), or moderate blemishes (≤20%).
Reject (Unsound)	0	0.0%	Severe visible rot, extensive surface decay (>20%), or major pathological defects.

3. Metrology & Physical Scale Calibration

Calibration State:	ESTIMATED	Ref Marker Detected:	YES (Localized in scene)
Known Reference Std:	25.0 mm standard	Scale Factor:	4.76 px/mm (0.2101 mm/px)

CALIBRATED PHYSICAL MEASUREMENT: Metric millimetre values are calculated using verified planar reference calibration (4.76 px/mm). Absolute measurements are valid under co-planar camera positioning.

4. Computer Vision Segmentation Overlay



5. Individual Onion Detection & Morphometry Records

Detailed inspection record for each localized onion instance, including neural classification confidence, morphological parameters, and deterministic grading rationale.

	Bulb #1 — Red-Onion	Grade: GRADE C	Health: Healthy	Review: AUTO_ACCEPTABLE
	Seg Conf: 96.2%	Health Conf: 100.0%	Combined Conf: 98.1%	Defect Area: 0.0%
	PIXEL MEASUREMENT:	Eq. Diameter: 117.6 px Area: 10858 px² Perimeter: 441.2 px Circularity: 0.70 Aspect Ratio: 1.70		
	CALIBRATED MEASUREMENT:	Calibrated Diameter: 24.7 mm		
	Grading Rationale:	• Healthy classification with high confidence. • Calibrated size estimated at 24.7 mm. • High confidence segmentation and classification with consistent geometry.		

6. Technical Limitations & Engineering Disclaimers

- 1. Visible Surface Inspection:** RGB optical inspection captures only visible external outer tunic and epidermal features. Subsurface physiological decay, internal core rot, black mold beneath dry tunics, and bacterial soft rot cannot be evaluated without destructive cutting or non-visible spectral scanning.

2. Internal Rot Limitation: A 'Healthy' external surface classification does not guarantee the absence of interior fungal or bacterial degradation.

3. Scale Calibration Requirement: Absolute millimeter measurements require an explicit planar reference standard. When uncalibrated, only raw pixel morphometry is recorded; metric physical dimensions are never fabricated.

4. Prototype Grading Notice: Automated grading tiers (Grade A/B/C/Reject) are derived from an engineering prototype rule engine for Smart India Hackathon technical evaluation and decision support. They do not constitute statutory agricultural certification (such as AGMARK, NAFED, or export grade certificates).

5. Confidence Metric: Model confidence denotes neural network softmax certainty on visible visual patterns, not verified laboratory accuracy or shelf-life durability.

6. Decision Support: Results are intended for assisted inspection and preliminary screening. Critical commercial batches require certified laboratory metrology and expert human inspection.